

Raphaël GILLET

Senior Data Scientist

Production Machine Learning · MLOps · Python · SQL · AWS · Snowflake · Polyglot

Over 5 years delivering Machine Learning end-to-end — development, deployment and performance monitoring — for banking and marketing clients.

- **Impact:** 0% → 50%+ lift over control and 80k+ conversions a year (major US bank)
- **Production:** XGBoost, Bayesian tuning, daily drift and data-quality monitoring
- **Background:** Physics, coastal-erosion engineering · native French, C2 English & Italian

raphael.gillet1@gmail.com | +33 782 336 344 | Paris, France | Portfolio | GitHub

EXPERIENCE

Zeta Global — *Prague (2022 – 2024) · Paris (since 2025)*

Senior Data Scientist — February 2022 – Present

Banking client, one of the largest US financial institutions — Site AI Optimization engine (next-best-product recommendation), 2023 – Present

- Led the transformation of the Site AI Optimization engine — the next-best-product recommendation model on the client's website homepage — and improved its model lift over control from 0% to 50%+, which drove over 80k incremental conversions annually.
- Engineered the underlying Machine Learning model with XGBoost, ranking banking and card offers for prospects and existing customers.
- Rebuilt the hyperparameter tuning as a Bayesian search that runs through the exact production pipeline and evaluation, so tuned settings transfer directly to deployment — train/serve parity as a deliberate industrialization choice.
- Designed the post-launch performance monitoring, which reads live model scores from Snowflake (SQL) and flags drift, traffic and data-quality issues daily.
- Contributed to the model refresh and model governance process agreed with the client, supporting the engine's operation in a regulated financial-services environment.

Other Zeta engagements — B2B customer acquisition and web-page optimization

- Built and maintained automated Machine Learning pipelines for B2B customer-acquisition services, driving significant revenue.
- Collaborated on web-page optimization Machine Learning models for ad selection and placement, increasing customer conversions and revenue.
- Worked cross-functionally with data engineering to streamline the transition of models from development to production, translating models to PMML for live scoring.

Majorel — *Paris, France*

Data Scientist — 2021 – 2022

- Collected, analysed and modelled customer data, mostly for the banking and insurance sectors (Python, SQL).

French Geological Survey — *Orléans, France*

Junior Research Fellow in Coastal Engineering — 2017

- Modelled erosion-hazard indicators along the Aquitaine coastline, relating storm conditions to beach erosion as groundwork for a storm-erosion early-warning system.
- First author of the related paper at the International Coastal Symposium 2018 (Journal of Coastal Research), listed below.
- Contributed to the creation of a national network on erosion-related studies.

Richelieu High School *Rueil-Malmaison, France*

Physics & Chemistry Teacher 2016

- Taught physics and chemistry to high-school classes, explaining technical concepts to non-specialist audiences.

EDUCATION

MSc-level coursework in Data Science 2021

OpenClassrooms / CentraleSupélec — Online

MSc in Physical Geography 2017

Université Paris Diderot — Paris, France

BSc in Physics 2013

Université Pierre et Marie Curie / Sapienza Università di Roma — Paris / Rome

PROJECTS & PUBLICATIONS

Lab-of-origin attribution for engineered DNA

1-D convolutional network (TensorFlow / Keras) classifying 63,017 plasmid DNA sequences among 1,314 laboratories, with hyperparameters tuned by Bayesian optimization on validation top-10 accuracy. Top-10 accuracy of 0.764 on a held-out test set of 12,604 sequences.

Storm Impact Indicator — beach erosion vs. storm energy (Truc Vert, France)

Physics-based wave run-up indicator (Stockdon et al., 2006) built on WaveWatch III / MARS2D wave and water-level series and 162 beach surveys; regressions by prior beach state, evaluated on a chronological hold-out (RMSE 8.1 m^3/m).

Towards the development of a storm erosion Early Warning System for the French Aquitanian coast

Gillet R. et al. (2018). Proceedings of the International Coastal Symposium (ICS) 2018, Busan, Republic of Korea — Journal of Coastal Research.

A comparative hydro-morphometric approach for modelling rainfall variability impact during hydrological extremes: application to 2002 and 2003 catastrophic events in the Lower Rhône valley, France

Henaff Q., Arnaud-Fassetta G., Beltrando G., Gillet R. (2014). EGU General Assembly, Vienna, Austria.

SKILLS

Programming: Python, SQL, R, Bash, Git

Machine Learning: XGBoost, scikit-learn, TensorFlow/Keras; classification, regression, ranking / recommendation models, Bayesian hyperparameter optimization, automated ML pipelines

MLOps & Data Platforms: AWS (S3, EC2), Snowflake, Docker, GitLab CI/CD, Luigi (workflow orchestration), ONNX model serving, production pipelines

LLM tools: Claude, OpenClaw, Hermes

LANGUAGES

English (C2), French (native), Italian (C2), Spanish (B2), Portuguese (B2), Turkish (A1)

ADDITIONAL

Interests: chess (Elo ≈ 1900), sailing, meditation · Driving licence